

1. Introduction

Cloud computing allows users to access computing resources such as virtual machines, storage and applications over the internet instead of owning physical hardware. IBM describes cloud computing as “a model for delivering computing resources over the internet on demand” (IBM, 2024a). One major benefit is that users only pay for what they use, meaning the cloud enables flexibility, scalability and cost effectiveness for both small and large workloads.

In this lab, we compare the performance of two virtual machines (VMs) with different configurations. We run the same identical code on both machines and measure execution time and cost.

2. Computational Formulas and VM Performance Table

The values in the results table come from executing these two mathematical tasks:

1.

$$x = \sum_{i=1}^n (a_i^4 + b_i^2)^3 + \sum_{i=1}^m (a_i^2 - b_i)^2$$

2.

$$Y1 = A1.B1$$

$$Y2 = A2.B2$$

$$Y = Y1 + Y2$$

where each matrix is 600 x 600 with elements less than 100.

Below is the table summarizing the two Azure VMs used in this experiment.

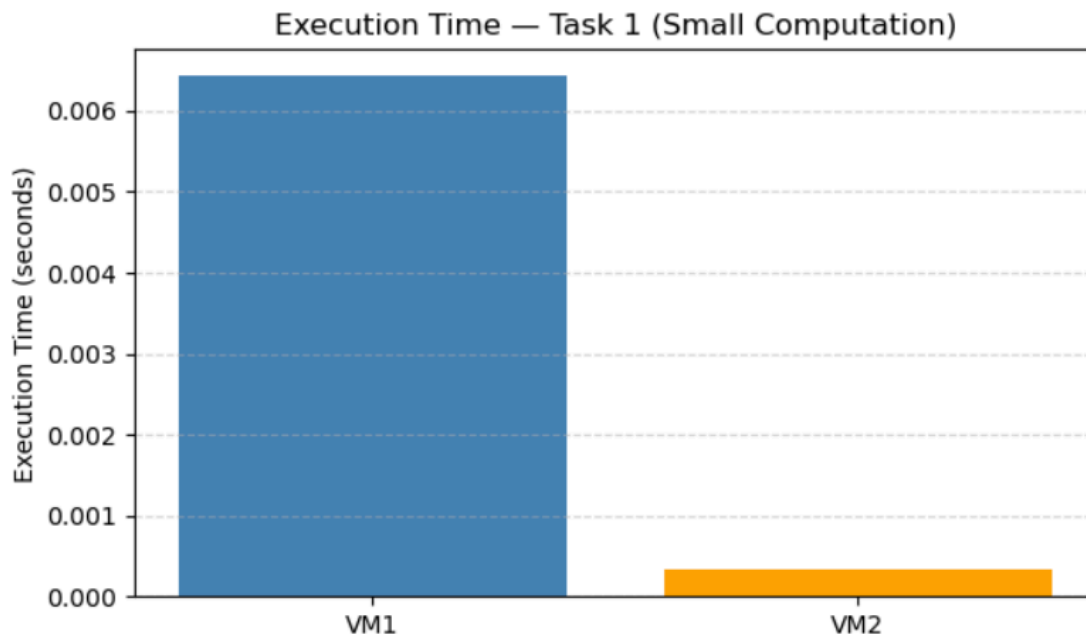
VM	Cores	RAM	Task 1 time (s)	Task 2 time (s)	Price/hour	Task 1 cost	Task 2 cost
VM1	2	4 GB	0.006436	1.922357	\$0.20	$0.006436/3600 \times 0.20 = \0.00000036	$1.922357/3600 \times 0.20 = \0.00010680
VM2	8	32 GB	0.000326	0.482052	\$0.84	$0.000326/3600 \times 0.84 = \0.00000008	$0.482052/3600 \times 0.84 = \0.00011240

Even though VM2 is more expensive per hour, the much faster execution of Task 1 makes its cost almost identical and even cheaper than VM1.

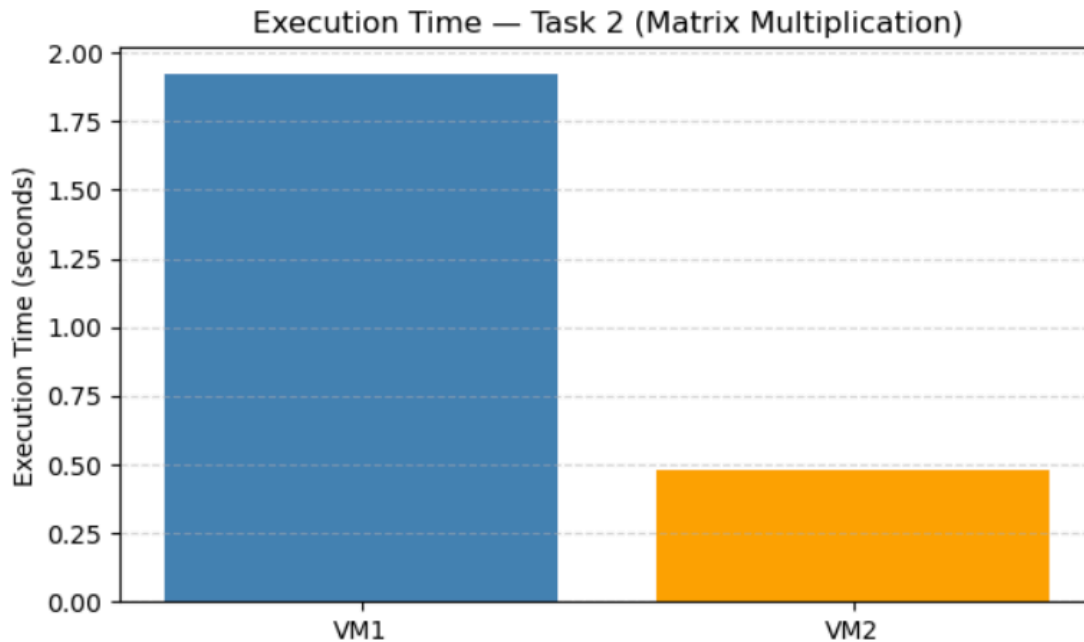
Task 2 is slightly more expensive on VM2, but it achieves dramatically better performance.

3. Results

3.1 Execution Time Comparison



VM2 runs the small task much faster than VM1.



VM2 completes the heavy task noticeably faster than VM1.

3.2 Performance Improvement

Using the formula:

$$\text{speedup} = \frac{TVM1}{TVM2}$$

$$\text{Task 1: } \frac{0.006436}{0.000326} \approx 19.75 \text{ x faster}$$

$$\text{Task 2: } \frac{1.922357}{0.482052} \approx 3.98 \text{ x faster}$$

Which means that:

Task 1: VM2 is ≈ 19.75 x faster

Task 2: VM2 is ≈ 3.98 x faster

This means VM2 can process both small formulas and large matrix multiplications much faster. Cloud performance always depends on the instance type one chooses, so two VMs can behave very differently even with the same code.

4. Discussion

The results clearly show how cloud hardware affects performance. As IBM explains, cloud resources can be “scaled up or down depending on demand” (IBM, 2024b). VM2 represents a more powerful and scalable option.

Performance perspective

- VM2 completed Task 1 nearly 20x faster.
- VM2 completed Task 2 almost 4x faster.

Cost perspective

Even though VM2 cost more per hour, the programs run so fast that:

- Task 1 is cheaper on VM2.
- Task 2 is only slightly more expensive but completed much faster.

This reflects the core idea of cloud pricing.

You pay for the time you use, not for “owning” the machine.

The cloud makes it possible to choose the correct machine for each workload. Faster VMs reduce waiting time and may even reduce cost. This lab demonstrates why cloud computing is considered flexible, scalable and cost efficient for both organizations and in this case students.

5. Conclusion

This lab showed that running the same code on two different cloud VMs produced significantly different performance outcomes. VM2, was much faster in both tasks. Although VM2 has a higher hourly cost, the actual cost per task remained very low and competitive due to shorter execution times.

This demonstrates the advantages of the cloud’s model that are offered, you can choose computing power based on your needs and only pay for the resource consumed. The cloud provides a balance between speed, flexibility and cost, which is why it has become the preferred computing model for modern applications.

References

IBM. (2024a). *What is cloud computing?* IBM Think.
<https://www.ibm.com/think/topics/cloud-computing>

IBM. (2024b). *Cloud computing benefits.* IBM Think.
<https://www.ibm.com/think/topics/cloud-computing-benefits>